Feature Engineering

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

TEACHER GUIDE · 1 OF 9

Feature Engineering

> Prepare Feature Engineering: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Create features available at **prediction** time

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Learning Curves

Without opening the old lesson, explain one key idea from Learning Curves. Give an example and a mistake that the idea helps you avoid.

Look for: Read **training**-size learning curves

Reference example, not the only acceptable wording: Teacher explanation: Mean(5+7+6)/3=6; range 7-5=2.

Reproducible Experiments

Without opening the old lesson, explain one key idea from Reproducible Experiments. Give an example and a mistake that the idea helps you avoid.

Look for: Record enough detail to reproduce a run

Reference example, not the only acceptable wording: Teacher explanation: Mean 70 percent; range 80-60=20 percentage points, across the three declared runs.

Hold Out Test Data

Without opening the old lesson, explain one key idea from Hold Out Test Data. Give an example and a mistake that the idea helps you avoid.

Look for: Freeze the full evaluation procedure before testing

Reference example, not the only acceptable wording: Teacher explanation: Accuracy 15/20, precision 3/5, recall 3/6. Report TN 12 FP 2 FN 3 TP 3 and blocked positive.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Length=12 cm, width=30 mm. Compute length/width using matching units. Explain handling for zero or missing width and where learned scaling is fitted.

Concept to explain

Feature engineering transforms raw **input**s into useful structure while preserving **prediction**-time **availability** and avoiding **leakage**.

Term	Meaning
transformation	rule converting raw values
derived feature	new input calculated from existing inputs
aspect ratio	width divided by length
feature pipeline	fixed ordered transformations

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

> Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach two 45-minute sessions. This week focuses on **feature engineering**. Model the worked example first, collect the new W 4 case independently before feedback, and use the fresh retry to check transfer.

Award credit for traceable reasoning and bounded claims, including justified uncertainty.

Readiness answer

Evaluation values must not influence the learned **transformation**.

Printed student lesson and workbook

Paper, pencils and a calculator for fractions

The fictional examples and a blank **evidence** table

Saved classroom project records for critique; no personal student dataset required

Apply the connected idea: Relationships and confounding

Teach Week 21 as a two-session investigation using fictional data. In a fictional log, groups that practiced longer also scored higher: 1 hour with 4 points, 2 hours with 6 and 3 hours with 8. This pattern is an **association**.

Prior experience could affect both practice choices and scores. The table alone cannot establish the effect of adding practice time.

Assess the three stated outcomes through the independent task, not by copying the worked result.

Readiness answer

No; selection and coverage limit the claim.

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional **input** records from this lesson; no real student identifiers

End goal for the pictorial case

By the end, I can explain **feature engineering** using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model Feature Engineering using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Trace the information, fitted behavior, and evaluation proof separately.

Question to explain

QUESTION Create and justify two plant measurement features.

MODEL IDEA **Feature engineering** transforms raw **input**s into useful structure while preserving **prediction**-time **availability** and avoiding **leakage**.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD Raw leaf length=8 cm and width=4 cm. Derived aspect_ratio = 4 ÷ 8 = 0.5.

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

Step 1

STEP 1 Raw leaf length=8 cm and width=4 cm.

Step 2

STEP 2 Derived aspect_ratio = 4 ÷ 8 = 0.5.

Step 3

STEP 3 Color histogram summarizes pixel ranges without using the target label.

Step 4

STEP 4 Fit **transformation** settings on **training** data only; document unit, purpose, **availability**, and **leakage** check.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

Concrete teaching case

The teaching case

Features are leaf length and width in cm; add length/width.

The teaching case

Compute for8 and2; state boundary checks.

Exact explanation to model

Why the result follows

Ratio=4, unitless with matching units. Guard zero/missing width; fit learned transforms on **training** only.

Explain the added worked example

Explain the added worked example · 1

A useful feature represents information available when the **prediction** is made. In a fictional parcel task, length and width are measured before packing, so area=length*width is available.

The final delivery delay is not available when predicting dispatch time. A strong correlation cannot make a future value legitimate.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines **feature engineering**.

Training rectangles have lengths [2, 4, 6] and widths [3, 2, 1], so areas are[6, 8, 6] square cm. **Training** mean area is 20/3. A validation rectangle length 5 width 2 has area 10 and centered area 10-20/3=10/3.

Freeze the **training** mean; do not recompute it with validation.

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

Keep the distinction between an observed result and a proposed next step.

Key terms

Term	Meaning
Feature engineering	Constructing model inputs from available measurements
Derived feature	An input computed from other inputs
Leakage	Information entering a model that would not be available legitimately
Availability	Whether a value exists when a prediction is needed

Apply the connected idea: Relationships and confounding

Apply the connected idea: Relationships and confounding · 2

In a fictional log, groups that practiced longer also scored higher: 1 hour with 4 points, 2 hours with 6 and 3 hours with 8. This pattern is an **association**. Prior experience could affect both practice choices and scores.

The table alone cannot establish the effect of adding practice time.

Apply the connected idea: Relationships and confounding · 3

The numbers rise together in these three invented records. A description such as higher practice hours went with higher scores is supported. A claim that one extra hour guarantees two more points goes beyond the **observation**s.

No statistical calculation is required to see this limit.

Prior experience, motivation or available support could plausibly affect both. Explain the connection to each.

Exit: Small size, constructed data, conditions and measured outcome; do not generalize broadly or claim classroom effectiveness.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Feature engineering at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Raw measurements	Length and width	Available before prediction
Derived area	Multiply length by width	Units square cm
Centering	Subtract training mean	Reuse frozen value

Area is a deterministic transformation; it adds a representation, not a new measured fact. It may help a limited model express a relationship, or may do nothing useful. Compare original versus extended features under the same split, metric and development policy.

Center the validation area using the frozen training mean. Explain the steps or evidence behind your response.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Feature Engineering

Engineer a feature with a **transformation** defined before evaluating a new leaf.

The facts we will use

Supplied facts

Training rectangular leaf cards have areas [2, 6, 10] cm². Area=length×width; center by **training** mean. Query length 2, width 4 cm.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/29

Feature Engineering

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Construct area

Query area=2 x 4=8 cm2.



2. Fit centering

Training mean=(2+6+10)/3=6 cm2.



3. Apply unchanged transform

Centered query feature=8-6=2 cm2.

CHANGE ONE THING

Include the query in the centering mean.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Construct area	Query area=2×4=8 cm².
2. Fit centering	Training mean=(2+6+10)/3=6 cm².
3. Apply unchanged transform	Centered query feature=8-6=2 cm².

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. Centered query feature=8-6=2 cm². The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

Area is a geometric toy feature. Useful predictive relationships need separate evaluation, and real leaves need a suitable measurement definition.

Change one thing in the pictured case

Change one thing

Change: Include the query in the centering mean.

Predicted result

Prediction: Mean becomes26/4=6.5 cm² and query feature1.5 cm²; this changes the fitted **transformation**.

Why the result follows

Reason: A held-out **input** should not silently redefine a **training**-fitted pre**process**ing step.

Ask while showing the picture

Ask while showing the picture · 4

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Feature Engineering, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: Features are leaf length and width in cm; add length/width.

2. State the decision to explain

Write the question to explain: Compute for8 and2; state boundary checks.

3. Reveal the worked reasoning

Reveal after **prediction**s: Ratio=4, unitless with matching units. Guard zero/missing width; fit learned transforms on **training** only.

Hold back the independent task

Keep for an independent attempt: A feature is leaf_length divided by leaf_width. A row has width zero. What should the pre**process**ing rule do before division?

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the

result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided response: Area 10; centered 10-20/3=10/3 square cm. Validation does not change the fitted mean.

Exit response: Compare with and without area on the same validation records, **fitting** all **transformation**s on **training** and reporting the declared metric and feature **availability**.

Case-specific teaching sequence

Stage · Teacher action

> **Stage:** Session 1 start to 7 minutes
>
> **Teacher action:** Use the prerequisite question: Why must preprocessing be fitted on training only? Check the explanation before proceeding.

> **Stage:** 7 to 18 minutes
>
> **Teacher action:** Read the opening and model the worked evidence. Ask students to identify feature engineering in context.

> **Stage:** 18 to 32 minutes
>
> **Teacher action:** Complete W 1 and W 2 with support: compute features; transform validation. Require the evidence, not only the final response.

> **Stage:** 32 to 45 minutes

> **Teacher action:** Read the visual row by row. Collect the guided explanation and compare reasoning in pairs. Connect each conclusion to the stated evidence.

> **Stage:** Session 2 start to 8 minutes
>
> **Teacher action:** Retrieve the distinction between feature engineering and derived feature. Invite a counterexample to an overconfident claim.

> **Stage:** 8 to 20 minutes
>
> **Teacher action:** Discuss the application and complete W 3: Why is delivery delay invalid for a prediction made before dispatch?

> **Stage:** 20 to 35 minutes
>
> **Teacher action:** Collect W4 independently before giving hints. Assess these outcomes: Create features available at prediction time; Fit feature transformations using training evidence only. Record any support used.

> **Stage:** 35 to 45 minutes
>
> **Teacher action:** Use W 5 and the exit response to assess limits. Give targeted feedback, then the fresh retry. Save evidence of independent understanding.

Teach the connected idea

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and explain the purpose in the opening section.
7 to 20 minutes	Read the worked example and diagram; ask learners to trace each step aloud.
20 to 33 minutes	Complete W1 and W2, then compare predictions with observed results.
33 to 45 minutes	Collect the guided prediction before running; discuss a mismatch and its cause.
Session 2 start to 8 minutes	Retrieve the main rule without showing the previous answer.

Stage	Teacher action
8 to 20 minutes	Use W3 to diagnose and repair the stated failure; rerun an earlier case.
20 to 35 minutes	Collect W4 independently, including code or plan, predictions and explanations, before hints.
35 to 45 minutes	Discuss W5 and the exit; use the fresh retry when support was needed.

Reduce help deliberately

Completion practice

Completion task: Center the validation area using the frozen **training** mean.

Reasoning to listen for

Teacher check: Teacher explanation: Area 10; centered 10-20/3=10/3 square cm. Validation does not change the fitted mean.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for **Feature engineering**. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: Ask for the relevant counts, units, procedure or saved record. For this

week: Areas 6, 8, 6 square cm; **training** mean 20/3 square cm.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Connected teaching cycle: Relationships and confounding

Target: Describe **association**s without claiming **cause**

Reserve further sessions for this topic; the 135-minute primary plan does not absorb all added content.

Completion practice: Explain how prior experience could relate to both practice time and score.

Reasoning to listen for

Teacher check: Experienced learners may choose different practice amounts and also score higher even without additional practice.

Individual check: For Relationships and confounding, use its worked case to evaluate this proposed explanation: “A successful run proves the conclusion”. Explain the deciding step.

Expected repair

Expected correction: A fairer **comparison** could offer the same task, instructions and assessment and randomly assign fictional groups to practice conditions.

In real classrooms, a teacher must approve any activity and protect equal learning opportunities; this lesson uses a paper plan.

Apply the same support, independent transfer and delayed-review decisions to the named connected-topic workbook tasks.

Pictorial practice question

What should be saved so another group transforms the query identically? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Reasoned teacher answer

The area definition, cm² units and **training** mean 6; the correct declared centered feature is 2 cm². Changed-case check: Mean becomes26/4=6.5 cm² and query feature1.5 cm²; this changes the fitted **transformation**.

A held-out **input** should not silently redefine a **training**-fitted pre**process**ing step.

Changed-case reasoning: Mean becomes26/4=6.5 cm² and query feature1.5 cm²; this changes the fitted **transformation**. A held-out **input** should not silently redefine a **training**-fitted pre**process**ing step.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

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Run the weekly evidence lab

> Run the Feature Engineering investigation and use the first unsupported step to choose a teaching response.

Keep the test conditions fixed, record every result, and change only one planned factor.

1. Write the hypothesis and expected **evidence** before running.

2. Feature Workshop: design two **transformation**s, compute examples, audit **leakage**, and

compare one controlled validation result.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

MISCONCEPTION TO REPAIR Creating many features automatically gives the model more useful information.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A final answer is enough: Ask for the relevant counts, units, procedure or saved record. For this week: Areas 6, 8, 6 square cm; **training** mean 20/3 square cm.

A stronger claim makes a stronger project: Compare with and without area on the same validation records, **fitting** all **transformation**s on **training** and reporting the declared metric and feature **availability**.

The new case can be copied from the example: The independent task changes the **evidence**. Require a fresh explanation: **Training** mean 8 square cm; new area 12 square cm; centered feature 4 square cm.

Check these misconceptions

A successful run proves the conclusion: A fairer **comparison** could offer the same task, instructions and assessment and randomly assign fictional groups to practice conditions.

In real classrooms, a teacher must approve any activity and protect equal learning opportunities; this lesson uses a paper plan.

Matching the example is enough: Require the new W4 case, explicit **prediction**s and an explanation. Fresh retry: a log links more library visits with higher reading scores. Reading interest may influence both.

Require a supported **association** statement and a fairer **comparison** plan.

A convenient shortcut preserves meaning: A chart title Practice **cause**s success overstates the **evidence**. Replace it with Practice time and scores in three fictional records, and describe the possible **confounder**.

Better wording does not repair the study, but it makes the report honest.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Feature Engineering questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Ratio=4, unitless with matching units. Guard zero/missing width; fit learned transforms on

training only.

Independent question

Independent question · withhold the key

A feature is leaf_length divided by leaf_width. A row has width zero. What should the pre**process**ing rule do before division?

Expected independent answer

Teacher key · Expected independent answer

Detect zero or invalid width, investigate or handle it using a documented rule, and avoid division by zero. Apply pre**process**ing consistently without learning from test data.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “A feature formula is safe for every row just be**cause** it works on typical examples.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

Width=3 cm, ratio=4 and unitless. Zero width makes division undefined; missing width is not zero. Use a documented validation/fallback policy.

Fit learned scaling only on **training** data and apply it unchanged to other partitions.

Explanations for the additional workbook examples

W1 Compute features

Workbook question · W1

Compute the worked areas and their **training** mean.

Teacher reasoning · W1

Teacher explanation: Areas 6, 8, 6 square cm; **training** mean 20/3 square cm.

W2 Transform validation

Workbook question · W2

Center the validation area using the frozen **training** mean.

Teacher reasoning · W2

Teacher explanation: Area 10; centered 10-20/3=10/3 square cm. Validation does not change the fitted mean.

W3 Reject leakage

Workbook question · W3

Why is delivery delay invalid for a **prediction** made before dispatch?

Teacher reasoning · W3

Teacher explanation: The value is only known later, so using it would not match the information available at **prediction** time.

W4 Apply independently

Workbook question · W4

Training areas [4, 8, 12]. A new rectangle has length 3 width 4. Compute the **training** mean and centered new feature; state units.

Teacher reasoning · W4

Teacher explanation: **Training** mean 8 square cm; new area 12 square cm; centered feature 4 square cm.

W5 Plan a fair comparison

Workbook question · W5

How would you test adding area without claiming it must help?

Teacher reasoning · W5

Teacher explanation: Compare with and without area on the same validation records, **fitting** all **transformation**s on **training** and reporting the declared metric and feature **availability**.

Relationships and confounding — W1 Describe the evidence

State the pattern in the three worked records without using **cause**d or guaranteed.

Higher recorded practice hours are associated with higher scores in these three records.

Relationships and confounding — W2 Identify a confounder

Explain how prior experience could relate to both practice time and score.

Experienced learners may choose different practice amounts and also score higher even without additional practice.

Relationships and confounding — W3 Repair an overclaim

A learner predicts every student gains 2 points per extra hour. Explain why that is unsupported.

The three **observation**s do not establish a universal causal rule or predict every individual; other factors and variation are unmeasured.

Relationships and confounding — W4 Plan independently

A fictional dataset links more hints used with lower scores. Give a noncausal explanation and plan a fairer **comparison** of whether hints help.

Learners finding tasks harder may use more hints and score lower.

Use comparable tasks and starting skill, a teacher-approved or paper random-assignment plan, same outcome measure and report uncertainty; do not deny needed support.

Relationships and confounding — W5 State a remaining limit

> Even a better **comparison** has only four fictional groups. What should be reported?

Small size, constructed data, conditions and measured outcome; do not generalize broadly or claim classroom effectiveness.

Assess independent transfer

> **Independent question · withhold the key**
>
> Selected case: **Training** areas [4, 8, 12]. A new rectangle has length 3 width 4. Compute the **training** mean and centered new feature; state units.

Selected case answer

> **Teacher key · Selected case answer**
>
> Teacher explanation: **Training** mean 8 square cm; new area 12 square cm; centered feature 4 square cm.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Feature Engineering to Responsible Plant Classifier, adapt support and plan a later changed-case check.

Responsible Plant Classifier

Add only justified Plant Classifier features, with formula, source fields, expected benefit, and **leakage** test.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 30: Improve Generalization. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Provide a blank **evidence** table and allow a verbal explanation before writing. Highlight the given conditions without supplying the W 4 conclusion. Use **feature engineering** and **derived feature** in full sentences.

Extension: alter one condition in the worked example and explain which conclusion changes and which remains supported.

For independent W 4, require: **Training** mean 8 square cm; new area 12 square cm; centered feature 4 square cm. Fresh retry **training** areas [3, 6, 9], new rectangle 2 by 5: **training** mean 6, new area 10, centered 4 square units.

Require **training**-only **fitting**. Mark each criterion as independent, supported or not yet demonstrated. Reassess the reasoning after feedback; do not replace the first attempt silently.

Next week: Experimental records and seeds. Carry forward the saved **evidence** and the distinction between development choices and justified claims.

Readiness and reassessment for Relationships and confounding

Draw arrows from the proposed **confounder** to both observed variables. Extension: explain why reversing **cause** and effect can also fit some **observation**al patterns.

Independent W4 **evidence**: Learners finding tasks harder may use more hints and score lower.

Use comparable tasks and starting skill, a teacher-approved or paper random-assignment plan, same outcome measure and report uncertainty; do not deny needed support.

Record whether the learner needed no help, a hint or explanation. Fresh retry: a log links more library visits with higher reading scores. Reading interest may influence both.

Require a supported **association** statement and a fairer **comparison** plan.

Plan the delayed return

Suggested return points in this level: ai-4/30, ai-4/32, ai-4/35. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Feature Engineering lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Create features available at **prediction** time

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Center the validation area using the frozen **training** mean.

Supported practice answer

Reasoned practice answer: Teacher explanation: Area 10; centered 10-20/3=10/3 square cm. Validation does not change the fitted mean.

Check every learner before discussion: Use the worked case for **Feature engineering**. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Ask for the relevant counts, units, procedure or saved record. For this week: Areas 6, 8, 6 square cm; **training** mean 20/3 square cm.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Include the query in the centering mean. Require a **prediction** and a reason before comparing. Expected result: Mean becomes26/4=6.5 cm² and query feature1.5 cm²; this changes the fitted **transformation**.

Reason: A held-out **input** should not silently redefine a **training**-fitted pre**process**ing step.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

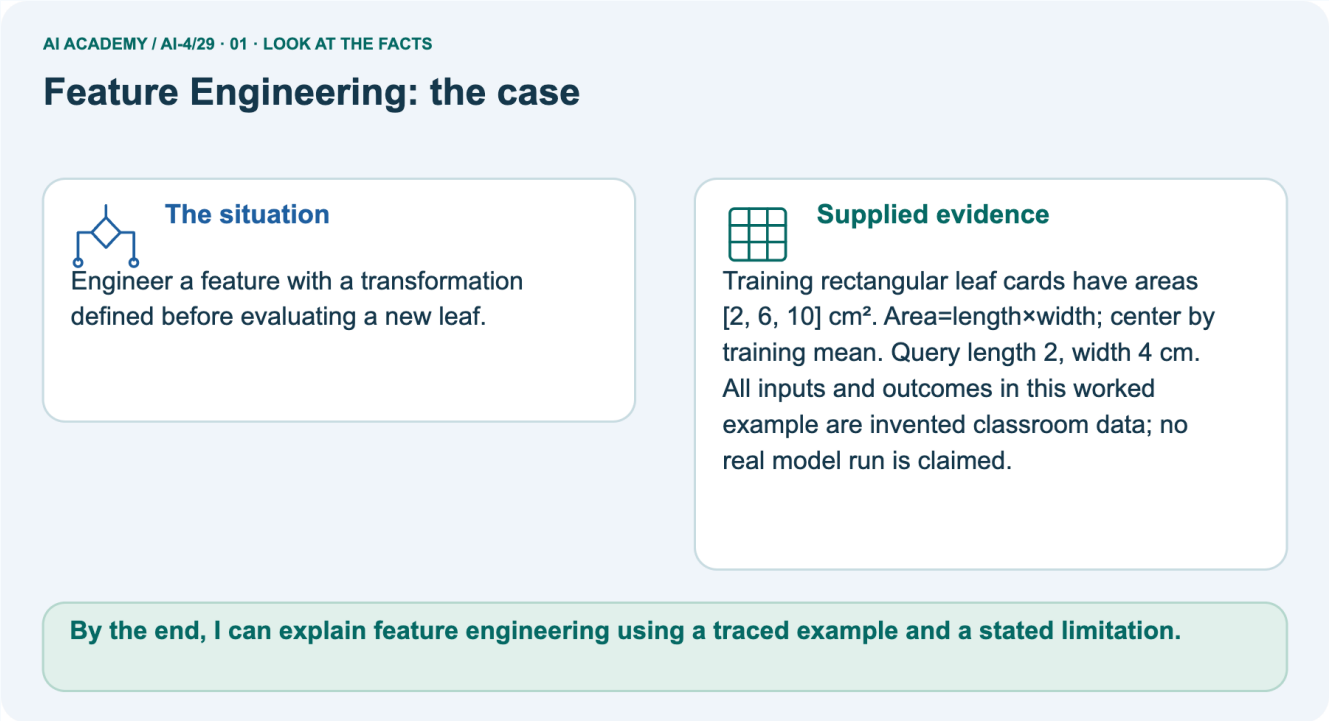
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Feature Engineering graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Feature Engineering: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

Engineer a feature with a **transformation** defined before evaluating a new leaf. Supplied facts: **Training** rectangular leaf cards have areas [2, 6, 10] cm². Area=length×width; center by **training** mean. Query length 2, width 4 cm.

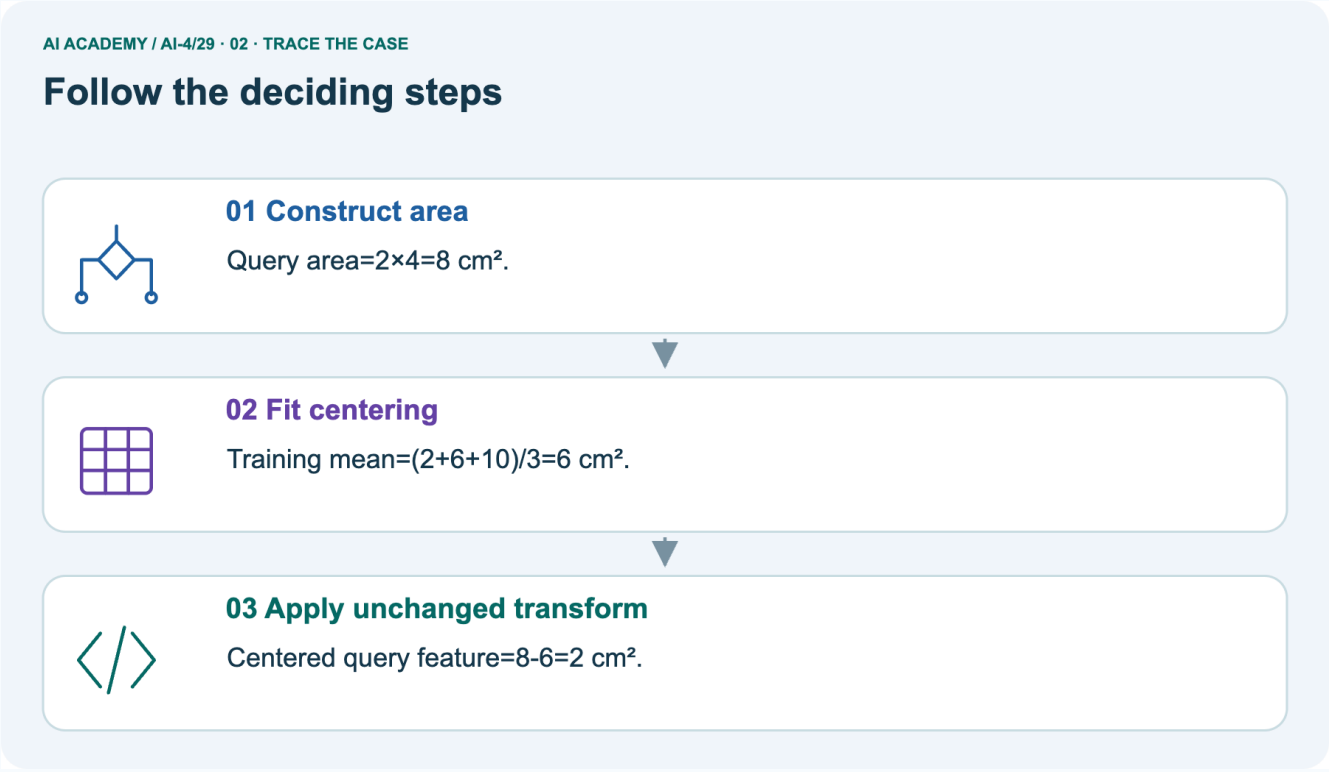
All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

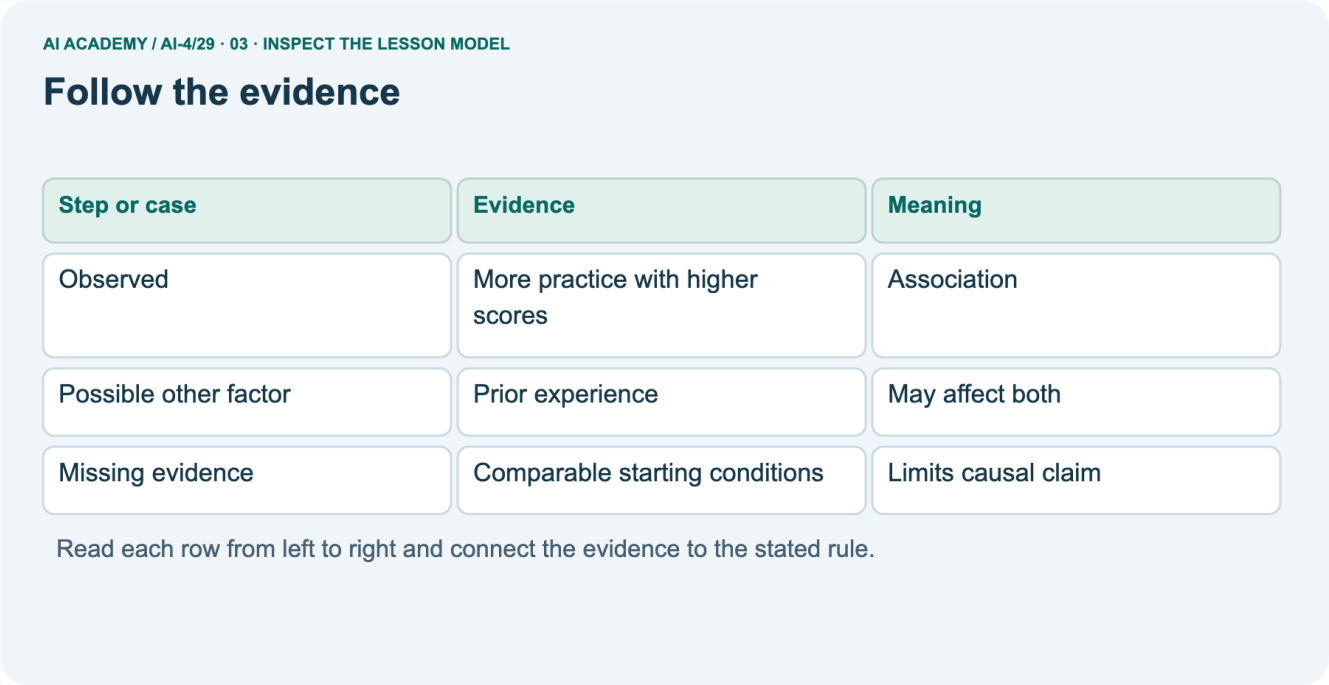
Construct area: Query area=2×4=8 cm². → Fit centering: **Training** mean=(2+6+10)/3=6 cm². → Apply unchanged transform: Centered query feature=8-6=2 cm².

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Follow the evidence



Follow the labelled picture with the matching explanation. Open the vector diagram

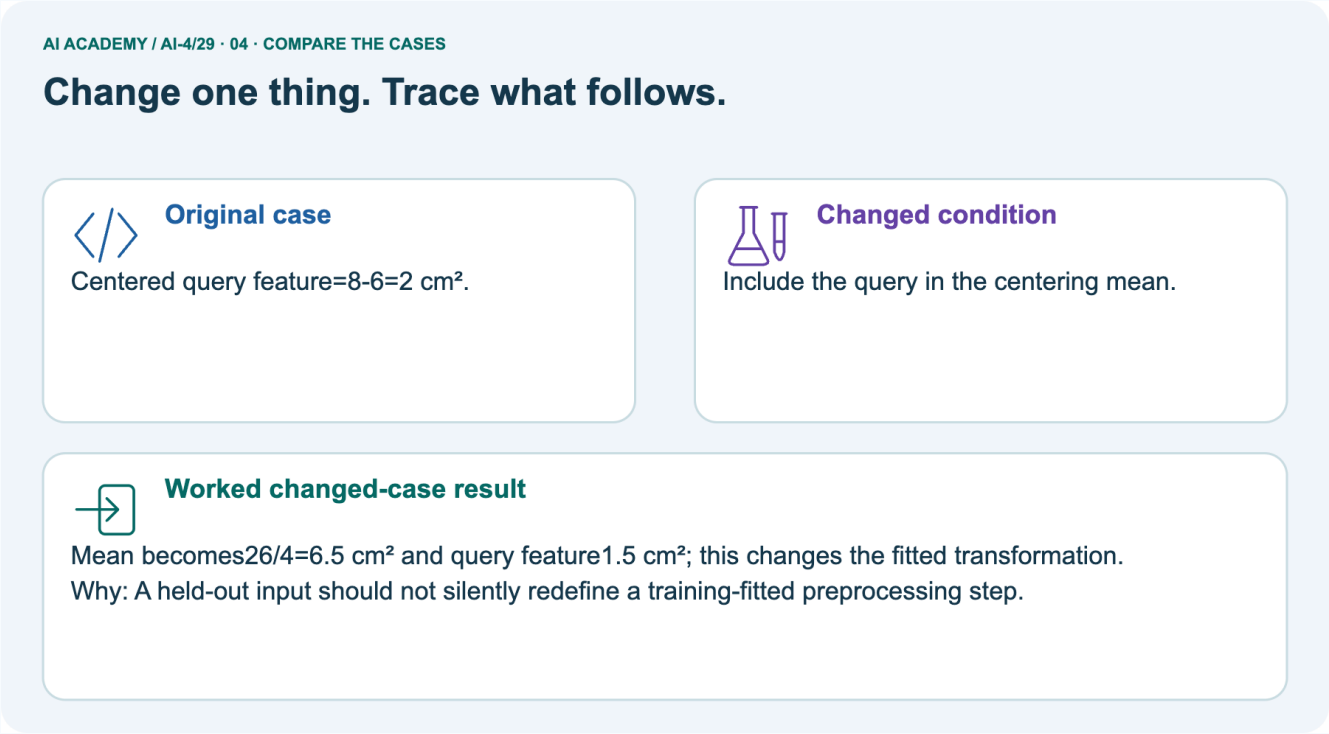
Follow the **evidence**. Observed | More practice with higher scores | **Association**. Possible other factor | Prior experience | May affect both. Missing **evidence** | Comparable starting conditions | Limits causal claim

Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Centered query feature=8-6=2 cm². Changed condition: Include the query in the centering mean. Expected result: Mean becomes26/4=6.5 cm² and query feature1.5 cm²; this changes the fitted **transformation**.

Reason: A held-out **input** should not silently redefine a **training**-fitted pre**process**ing step.

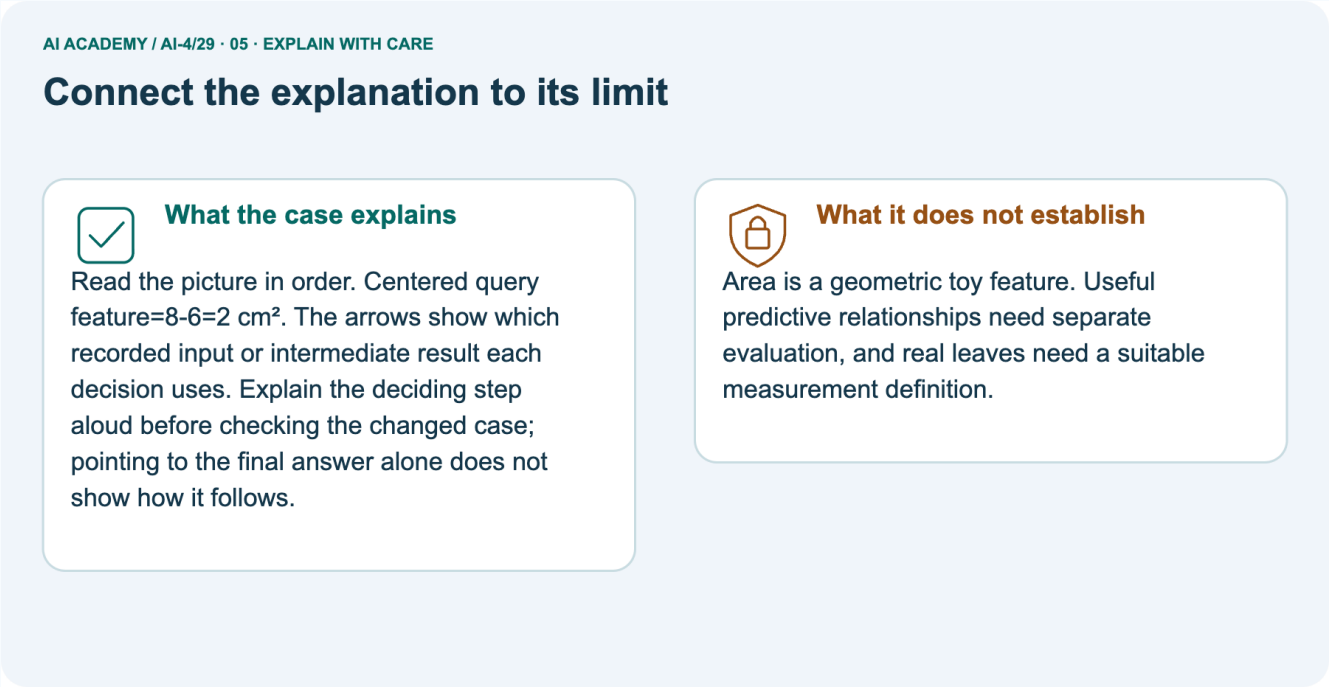
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: A held-out **input** should not silently redefine a **training**-fitted pre**process**ing step.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Centered query feature=8-6=2 cm². The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: Area is a geometric toy feature.

Useful predictive relationships need separate evaluation, and real leaves need a suitable measurement definition.

Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Area is a geometric toy feature. Useful predictive relationships need separate evaluation, and real leaves need a suitable measurement definition.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What should be saved so another group transforms the query identically? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The area definition, cm² units and **training** mean 6; the correct declared centered feature is 2 cm².

Changed-case check: Mean becomes26/4=6.5 cm² and query feature1.5 cm²; this changes the fitted **transformation**. A held-out **input** should not silently redefine a **training**-fitted pre**process**ing step.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.